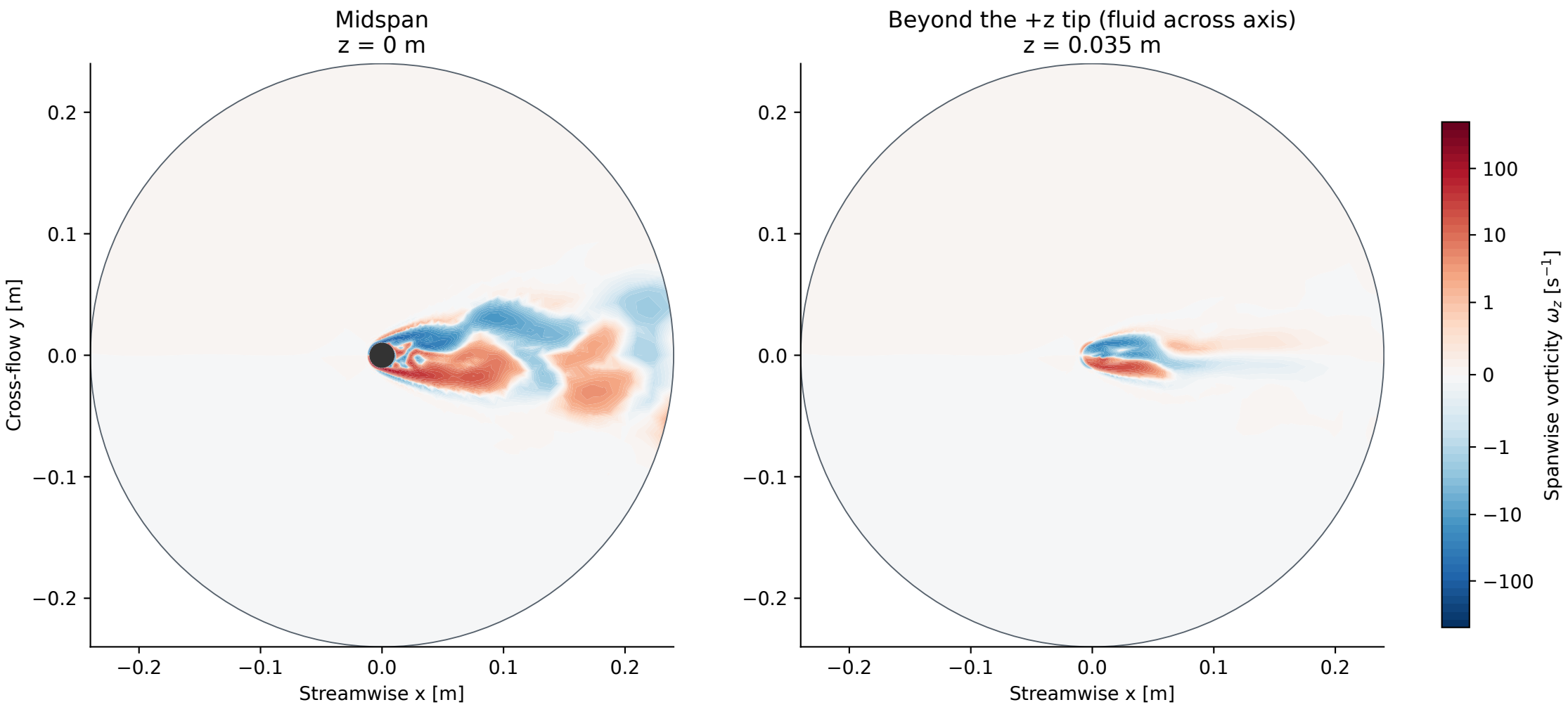

Circular-domain cylinder flow | t = 10 s
Full sampled domain: R = 0.24 m | Cylinder: D = 0.02 m, z = −0.03 to +0.03 m
Midspan
z = 0 m
Beyond the +z tip (fluid across axis)
z = 0.035 m
Cross-flow y [m]
Streamwise x [m]
Streamwise x [m]
Spanwise vorticity ω_z [s^{-1}]
0.2
0.1
0.0
−0.1
−0.2
100
10
1
0
−1
−10
−100
Sample-based linear contours; no extrapolation through solid or beyond sampled support.
Shared symmetric-log colour scale (linear within ±1 s⁻¹); scale limits ±499.740 s⁻¹. LES demonstration, not validation.